

The Ephemeris



March 2022

Volume 33 Number 01- The Official Publication of the San Jose Astronomical Association

March 2022 - June 2022 Events

Schedule subject to change due to Covid-19 restrictions.
Please check our website for up to date information.

Board & General Meetings (online)

Saturday 3/19, 4/23, 5/22, 6/26

Board Meetings: 6 -7:30pm

General Meetings: 8:00-9:30pm

Fix-It Day (2-4pm)

Currently on hold due to COVID-19 Shelter in Place

Solar Sunday (1:30-3:30pm)

Currently on hold due to COVID-19 Shelter in Place

Intro to the Night Sky Class &

Houge Park 1Q In-Town Star Party

Currently on hold due to COVID-19 Shelter in Place

Houge Park 3Q In-Town Star Party

Currently on hold due to COVID-19 Shelter in Place

RCDO Starry Nights Star Party

Currently on hold due to COVID-19 Shelter in Place

Imaging SIG

Tuesday 3/15, 4/21, 5/19, 6/16

Astro Imaging Clinics

(at Little Uvas OSP)

Currently on hold due to COVID-19 Shelter in Place

Binocular Star Gazing

Saturday (will resume in Summer 2022)

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

INSIDE THIS ISSUE

Annual Awards Board Elections	2
Open Positions	3
Articles	4
Astro Photos	7
Article	11
Kid Spot	12
Board of Directors	13
Club Programs	14
Membership Form	15

Flaming Star Nebula
Photo by SJAA Member: Steve Sells
January Jan 1, Jan 7, Jan 8 2022
Location: Menlo Park, CA

SJAA Annual Awards and Board Elections

Credit: Abishek

This year at the February 19th 2022 annual meeting, SJAA announced the results of the board elections. The meeting was held online due to COVID-19 related restrictions. Directors Rob Jaworski, Wolf Witt and Glenn Newell have been reelected into their positions, and Steven Nelson has been elected to the Board.



Ken Miura also announced that Peter Melhus was this year's SJAA Award winner! He has served with SJAA as interim secretary, assisted with the electoral voting process, and chaired the 24" Dob task group. Thank you Peter, for your generous commitment of time, energy and service in your role. From everyone at SJAA, we truly appreciate your efforts!

Ken also announced that the March meeting will be his last as President, but will continue as a Board Director for one additional year. Thank you Ken!

The board values everyone's resolve in keeping the SJAA name flying high. It thanks all its volunteers for their contributions, their skills in community leadership, and for their ability to be an inspiration for all members of the amateur astronomy community. A heartiest congratulations to all winners of the board election and awards, and a thank you to every valued member of SJAA for their continued support in making SJAA a fun place!

Call for Volunteers and Open Positions

Credit: Peter Melhus

On behalf of the SJAA, I would like to remind you that we are always looking for people who are interested in serving as volunteers, or on the Board of Directors of SJAA. We just had a successful Board of Directors election, and the results were announced on February 19, but we now write directly to all SJAA members and do hope that some of you will consider volunteering for an **'officer' position or as a 'program leader'** in SJAA, as we slowly see some light for a post-COVID-19 era. We really need your help!

We are currently looking for people to run for the **executive committee office**, i.e. President, Vice President, Secretary or Treasurer. The term for the officers is for one year. All officers, except the president, can run for reelection as often as desired. The president can serve for only two consecutive terms.

The **officer elections** are coming up on March 19, and the members of the Board of Directors will vote in this election. Three of the four positions above will have no incumbent running, and we only have the treasurer running for reelection. If you would like to be part of the executive team, now is the time to make your move. You'll find the duties of the officers (president, vice president, secretary and treasurer) described in our By-laws at <https://www.sjaa.net/about/club-by-laws/#am2> in Article 4, Sections 6-9.

And finally, we are looking for people to lead programs in which they have a specific interest. We have a number of **program leadership positions** open at this time. You can express your interest in leading one (or more) of these programs. The current vacancies we have are:

- Donations
- QuickSTART
- School Star Party
- Armchair

We need people to step up to help lead YOUR club! SJAA is YOUR organization. Only with your help will it continue to succeed and be a fun place!

To submit your name for consideration as an officer or a program leader, please email Peter Melhus (secretary@sjaa.net) with your name and a brief summary of your qualifications for the position you are interested in. Anyone interested in running for an officer position needs to contact the secretary by March 8.

How Dark Matter Could Be Measured in the Solar System

Credit: NASA

Pictures of the Milky Way show billions of stars arranged in a spiral pattern radiating out from the center, with illuminated gas in between. But our eyes can only glimpse the surface of what holds our galaxy together. About 95 percent of the mass of our galaxy is invisible and does not interact with light. It is made of a mysterious substance called dark matter, which has never been directly measured.

Now, a new study calculates how dark matter's gravity affects objects in our solar system, including spacecraft and distant comets. It also proposes a way that dark matter's influence could be directly observed with a future experiment. The article is published in the *Monthly Notices of the Royal Astronomical Society*.

"We're predicting that if you get out far enough in the solar system, you actually have the opportunity to start measuring the dark matter force," said Jim Green, study co-author and advisor to NASA's Office of the Chief Scientist. "This is the first idea of how to do it and where we would do it."

Dark matter in our backyard

Here on Earth, our planet's gravity keeps us from flying out of our chairs, and the Sun's gravity keeps our planet orbiting on a 365-day schedule. But the farther from the Sun a spacecraft flies, the less it feels the Sun's gravity, and the more it feels a different source of gravity: that of the matter from the rest of the galaxy, which is mostly dark matter. The mass of our galaxy's 100 billion stars is miniscule compared to estimates of the Milky Way's dark matter content.

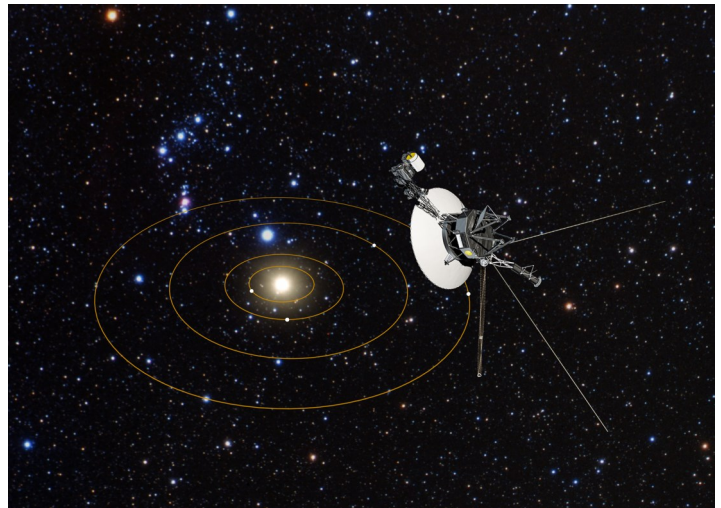
To understand the influence of dark matter in the solar system, lead study author Edward Belbruno calculated the "galactic force," the overall gravitational force of normal matter combined with dark matter from the entire galaxy. He found that in the solar system, about 45 percent of this force is from dark matter and 55 percent is from normal, so-called "baryonic matter." This suggests a roughly half-and-half split between the mass of dark matter and normal matter in the solar system.

"I was a bit surprised by the relatively small contribution of the galactic force due to dark matter felt in our solar system as compared to the force due to the normal matter," said Belbruno, mathematician and astrophysicist at Princeton University and Yeshiva University. "This is explained by the fact most of dark matter is in the outer parts of our galaxy, far from our solar system."

A large region called a "halo" of dark matter encircles the Milky Way and represents the greatest concentration of the dark matter of the galaxy. There is little to no normal matter in the halo. If the solar system were located at a greater distance from the center of the galaxy, it would feel the effects of a larger proportion of dark matter in the galactic force because it would be closer to the dark matter halo,

the authors said.

How dark matter may influence spacecraft



In this artist's conception, NASA's Voyager 1 spacecraft has a bird's-eye view of the solar system. The circles represent the orbits of the major outer planets: Jupiter, Saturn, Uranus, and Neptune. Launched in 1977, Voyager 1 visited the planets Jupiter and Saturn. The spacecraft is now more than 14 billion miles from Earth, making it the farthest human-made object ever built. In fact, Voyager 1 is now zooming through interstellar space, the region between the stars that is filled with gas, dust, and material recycled from dying stars.

Credits: NASA, ESA, and G. Bacon (STScI)

Green and Belbruno predict that dark matter's gravity ever so slightly interacts with all of the spacecraft that NASA has sent on paths that lead out of the solar system, according to the new study.

"If spacecraft move through the dark matter long enough, their trajectories are changed, and this is important to take into consideration for mission planning for certain future missions," Belbruno said.

Such spacecraft may include the retired Pioneer 10 and 11 probes that launched in 1972 and 1973, respectively; the Voyager 1 and 2 probes that have been exploring for more than 40 years and have entered interstellar space; and the New Horizons spacecraft that has flown by Pluto and Arrokoth in the Kuiper Belt.

Continued on Page 5

Continued from How Dark Matter ...

But it's a tiny effect. After traveling billions of miles, the path of a spacecraft like Pioneer 10 would only deviate by about 5 feet (1.6 meters) due to the influence of dark matter. "They do feel the effect of dark matter, but it's so small, we can't measure it," Green said.

Where does the galactic force take over?

At a certain distance from the Sun, the galactic force becomes more powerful than the pull of the Sun, which is made of normal matter. Belbruno and Green calculated that this transition happens at around 30,000 astronomical units, or 30,000 times the distance from Earth to the Sun. That is well beyond the distance of Pluto, but still inside the Oort Cloud, a swarm of millions of comets that surrounds the solar system and extends out to 100,000 astronomical units.

This means that dark matter's gravity could have played a role in the trajectory of objects like 'Oumuamua, the cigar-shaped comet or asteroid that came from another star system and passed through the inner solar system in 2017. Its unusually fast speed could be explained by dark matter's gravity pushing on it for millions of years, the authors say.

If there is a giant planet in the outer reaches of the solar system, a hypothetical object called Planet 9 or Planet X that scientists have been searching for in recent years, dark matter would also influence its orbit. If this planet exists, dark matter could perhaps even push it away from the area where scientists are currently looking for it, Green and Belbruno write. Dark matter may have also caused some of the Oort Cloud comets to escape the orbit of the Sun altogether.

Could dark matter's gravity be measured?

To measure the effects of dark matter in the solar system, a spacecraft wouldn't necessarily have to travel that far. At a distance of 100 astronomical units, a spacecraft with the right experiment could help astronomers measure the influence of dark matter directly, Green and Belbruno said.

Specifically, a spacecraft equipped with radioisotope power, a technology that has allowed Pioneer 10 and 11, the Voyagers, and New Horizon to fly very far from the Sun, may be able to make this measurement. Such a spacecraft could carry a reflective ball and drop it at an appropriate distance. The ball would feel only galactic forces, while the spacecraft would experience a thermal force from the decaying radioactive element in its power system, in addition to the galactic forces. Subtracting out the thermal force, researchers could then look at how the galactic force relates to deviations in the respective trajectories of the ball and the spacecraft. Those deviations would be measured with a laser as the two objects fly parallel to one another.

A proposed mission concept called Interstellar Probe, which aims to travel to about 500 astronomical units from the Sun to explore that uncharted environment, is one possibility for such an experiment.



Two views from Hubble of the massive galaxy cluster Cl 0024+17 (ZwCl 0024+1652) are shown. To the left is the view in visible-light with odd-looking blue arcs appearing among the yellowish galaxies. These are the magnified and distorted images of galaxies located far behind the cluster. Their light is bent and amplified by the immense gravity of the cluster in a process called gravitational lensing. To the right, a blue shading has been added to indicate the location of invisible material called dark matter that is mathematically required to account for the nature and placement of the gravitationally lensed galaxies that are seen.

Credits: NASA, ESA, M.J. Jee and H. Ford (Johns Hopkins University)

More about dark matter

Dark matter as a hidden mass in galaxies was first proposed in the 1930s by Fritz Zwicky. But the idea remained controversial until the 1960s and 1970s, when Vera C. Rubin and colleagues confirmed that the motions of stars around their galactic centers would not follow the laws of physics if only normal matter were involved. Only a gigantic hidden source of mass can explain why stars at the outskirts of spiral galaxies like ours move as quickly as they do.

Today, the nature of dark matter is one of the biggest mysteries in all of astrophysics. Powerful observatories like the Hubble Space Telescope and the Chandra X-Ray Observatory have helped scientists begin to understand the influence and distribution of dark matter in the universe at large. Hubble has explored many galaxies whose dark matter contributes to an effect called "lensing," where gravity bends space itself and magnifies images of more distant galaxies.

Astronomers will learn more about dark matter in the cosmos with the newest set of state-of-the-art telescopes. NASA's James Webb Space Telescope, which launched Dec. 25, 2021, will contribute to our understanding of dark matter by taking images and other data of galaxies and observing their lensing effects. NASA's Nancy Grace Roman Space Telescope, set to launch in the mid-2020s, will conduct surveys of more than a billion galaxies to look at the influence of dark matter on their shapes and distributions.

NASA Asteroid Tracking System Now Capable of Full Sky Search

Credit: NASA



From left to right: Sutherland ATLAS station during construction in South Africa. Credit: Willie Koorts (SAAO); Chilean engineers and astronomers installing the ATLAS telescope at El Sauce Observatory. Credit: University of Hawaii; Illustration of NASA's DART spacecraft and the Italian Space Agency's (ASI) LICIACube prior to impact at the Didymos binary system.

Credits: NASA/Johns Hopkins, APL/Steve Gribben; Illustration of the NEO Surveyor spacecraft.

The NASA-funded Asteroid Terrestrial-impact Last Alert System (ATLAS)—a state-of-the-art asteroid detection system operated by the University of Hawai'i (UH) Institute for Astronomy (IfA) for the agency's Planetary Defense Coordination Office (PDCO)—has reached a new milestone by becoming the first survey capable of searching the entire dark sky every 24 hours for near-Earth objects (NEOs) that could pose a future impact hazard to Earth. Now comprised of four telescopes, ATLAS has expanded its reach to the southern hemisphere from the two existing northern-hemisphere telescopes on Haleakalā and Maunaloa in Hawai'i to include two additional observatories in South Africa and Chile.

"An important part of planetary defense is finding asteroids before they find us, so if necessary, we can get them before they get us" said Kelly Fast, Near-Earth Object Observations Program Manager for NASA's Planetary Defense Coordination Office. "With the addition of these two telescopes, ATLAS is now capable of searching the entire dark sky every 24 hours, making it an important asset for NASA's continuous effort to find, track, and monitor NEOs."

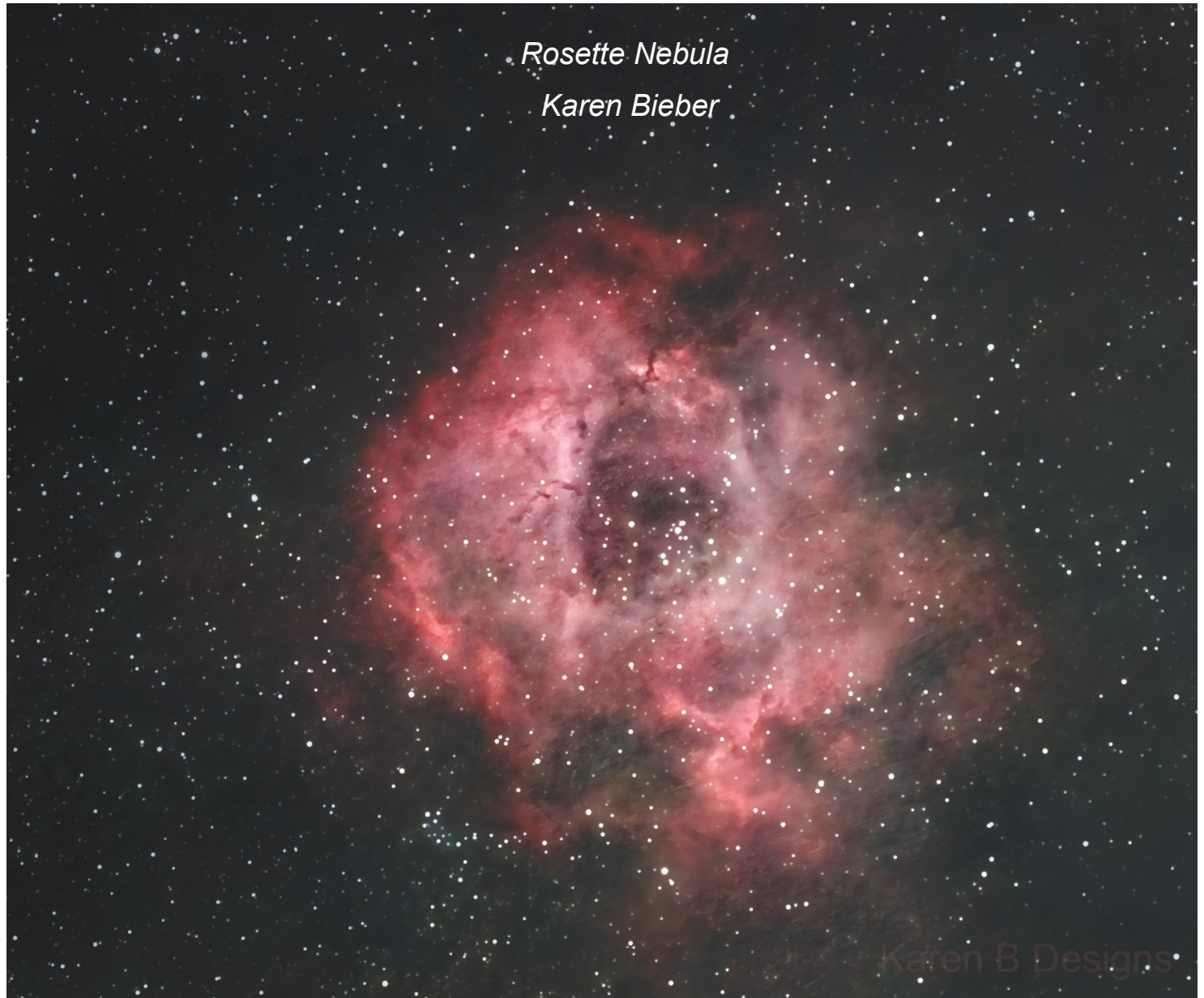
UH IfA developed the first two ATLAS telescopes in Hawai'i under a 2013 grant from NASA's Near-Earth Objects Observations Program, now part of NASA's PDCO, and the two facilities on Haleakalā and Maunaloa, respectively, became fully operational in 2017. After several years of successful operation in Hawai'i, IfA competed for additional NASA funds to build two more telescopes in the southern hemisphere. IfA sought partners to host these telescopes, and selected the South African Astronomical Observatory (SAAO) in South Africa and a multi-institutional collaboration in Chile. The ATLAS presence augments already substantial astronomical capability in both countries.

Each of the four ATLAS telescopes can image a swath of sky 100 times larger than the full moon in a single exposure. The completion of the two final telescopes, which are located at Sutherland Observing Station in South Africa and El Sauce Observatory in Chile, enable ATLAS to observe the night sky when it is daytime in Hawai'i.

To date, the ATLAS system has discovered more than 700 near-Earth asteroids and 66 comets, along with detection of 2019 MO and 2018 LA, two very small asteroids that actually impacted Earth. The system is specially designed to detect objects that approach very close to Earth - closer than the distance to the Moon, about 240,000 miles or 384,000 kilometers away. On January 22, ATLAS-Sutherland in South Africa discovered its first NEO, 2022 BK, a 100-meter asteroid that poses no threat to Earth.

The addition of the new observatories to the ATLAS system comes at a time when the agency's Planetary Defense efforts are on the rise. NASA's Double Asteroid Redirection Test (DART)—the world's first full-scale mission to test a technology for defending Earth against potential asteroid impacts—launched November 24, 2021 on a SpaceX Falcon 9 rocket from Space Launch Complex 4 East at Vandenberg Space Force Base in California. DART will deflect a known asteroid, which is not a threat to Earth, to slightly change the asteroid's motion in a way that can be accurately measured using ground-based telescopes.

Additionally, work on the agency's Near-Earth Object Surveyor space telescope (NEO Surveyor) is underway after receiving authorization to move forward into Preliminary Design, known as Key Decision Point- B. Once complete, the infrared space telescope will expedite the agency's ability to discover and characterize most of the potentially hazardous NEOs, including those that may approach Earth from the daytime sky.



Date: February 1, 2022

Location: Morgan Hill, Ca

Imaging Telescope: William Optics RedCat 71 Petzval APO

Imaging Camera: Canon Eos Ra

Mount: iOptron GEM45

Software: DSS DeepSkyStacker 2.4.6 · Adobe Photoshop CC 2021



Dates: November 4, 2021

Location: San Jose, CA

Telescope: Canon 70-200mm f/4 L IS USM

Camera: ZWO ASI 1600MM Pro

Mount: Orion Atlas AZ/EQ Pro

Software: NINA, PHD2, PixInsight

SJAA Member Astrophoto Gallery



Date: December 19, 2021

Location: San Francisco, CA

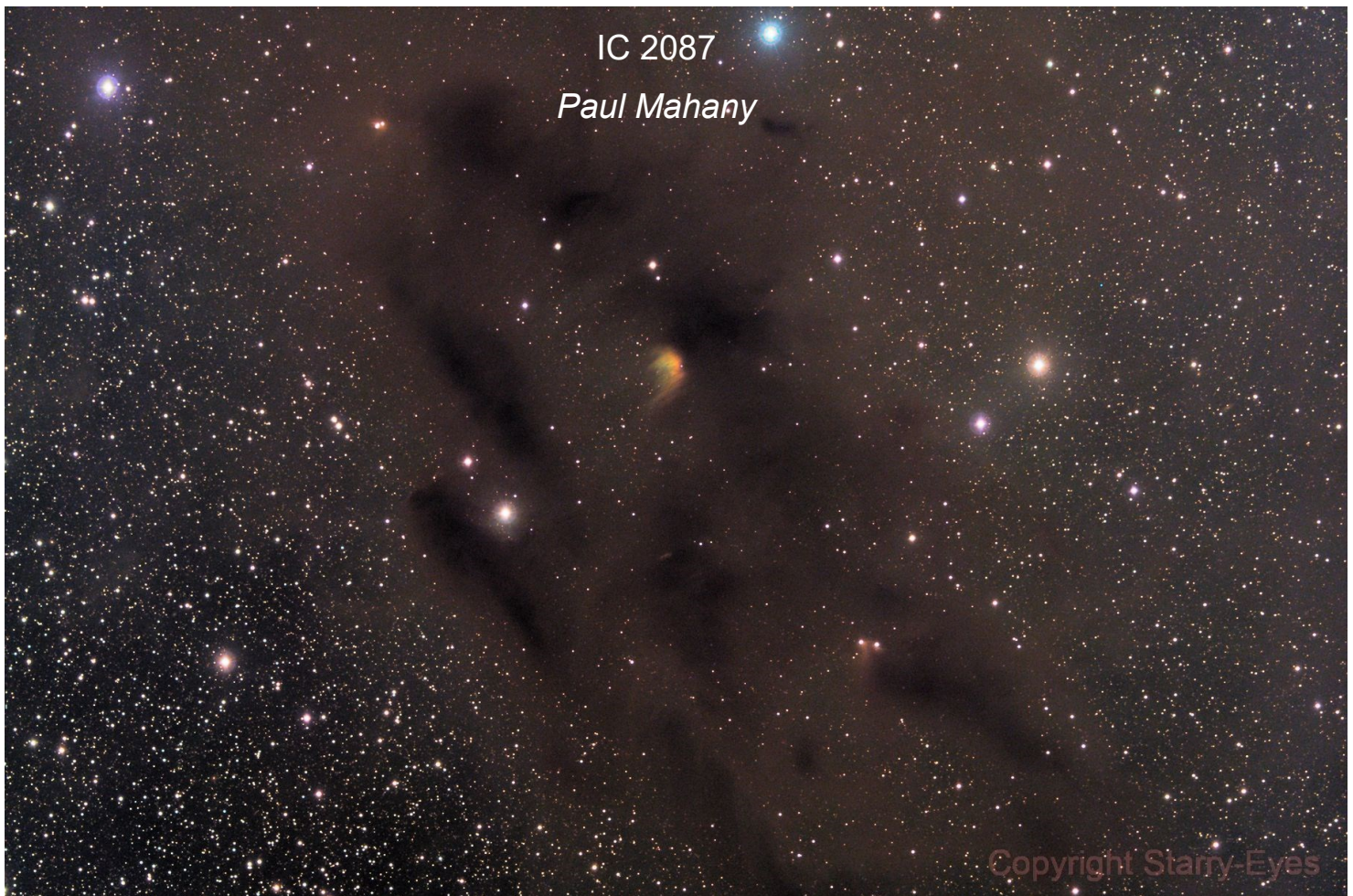
Telescope: Takahashi FSQ-106EDX4 Quadruplet Refractor

Camera: Nikon D800E

Mount: Astro-Physics Mach-1 GTO CP4

Software: Astro-Physics Inc. APCC-Pro, O'Telescope BackyardNikon, PixInsight 1.8, PHD2 Guiding

SJAA Member Astrophoto Gallery



Date: January 27, 2022

Location: Sierra Nevada Mountains, Near Yosemite, California, United States

Telescope: Celestron 11" Rowe-Ackermann Schmidt Astrograph

Camera: QHYCCD QHY268c

Mount: Software Bisque Paramount MX+

Software: SharpCap 3.1 Sharp Cap, StarTools, BYEOS, TheSkyX Pro, SkyTools, PixInsight1.8, PHD Guiding 2 PHD 2 Guiding

An Expanse of Light

Credit NASA



Credits: X-ray: NASA/CXC/SAO; Optical: NASA/STScI, Palomar Observatory, DSS; Radio: NSF/NRAO/VLA; H-Alpha: LCO/IMACS/MMTF

The recent launches of the James Webb Space Telescope (Webb) and the Imaging X-ray Polarimetry Explorer (IXPE) by NASA and its international partners are excellent reminders that the universe emits light or energy in many different forms. To fully investigate cosmic objects and phenomena, scientists need telescopes that can detect light across what is known as the electromagnetic spectrum.

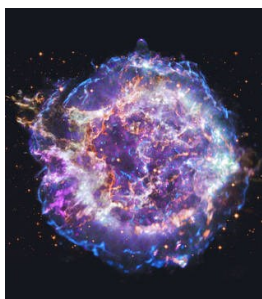
This gallery provides examples of the ways that different types of light from telescopes on the ground and in space can be combined. The common thread in each of these selections is data from NASA's Chandra X-ray Observatory, illustrating how X-rays — which are emitted by very hot and energetic processes — are found throughout the Universe.



R Aquarii:

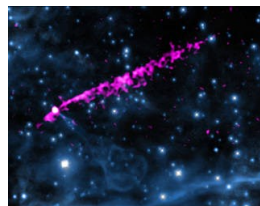
This object is, in fact, a pair: a white dwarf star that steadily burns at a relatively cool temperature and a highly variable red giant. As they orbit each other, the white dwarf pulls material from the red giant onto its surface. Over time, enough of this material accumulates and

triggers an explosion. Astronomers have seen such outbursts over recent decades. Evidence for much older outbursts is seen in the spectacular structures observed by NASA's Hubble Space Telescope (red and blue). X-ray data from Chandra (purple) shows how a jet from the white dwarf is striking material surrounding it and creating shock waves, similar to sonic booms from supersonic planes.



Cassiopeia A:

Chandra's observations of the Cassiopeia A supernova remnant have shown how individual elements from the exploded star are being cast off into space. In this image, X-rays reveal silicon (red), sulfur (yellow), calcium (green), and iron (light purple). The blue around the rim of the remnant reveals the blast wave from the explosion as it travels outward. This image also adds a layer of radio data of Cassiopeia A from the National Science Foundation's Karl Jansky Very Large Array (dark purple, blue, and white) and an optical image from Hubble (orange). Like X-rays, radio waves can penetrate thick clouds of gas and dust that lie between Earth and Cassiopeia A, providing additional information about this famous stellar explosion.



Guitar Nebula:

For the last decade or so, astronomers have been puzzled by the alignment of some jets of X-rays coming from very fast-moving pulsars (that is, spinning neutron stars) that shoot out into interstellar space at odd, unexpected angles. This is what astronomers see with PSR B2224+65, a pulsar found near the structure nicknamed the "Guitar Nebula" due to its shape in optical light (blue). An X-ray stream (pink) captured by Chandra is pointed nearly perpendicular to the guitar-shaped structure, originating from the magnetic poles of the pulsar.



Abell 2597:

Galaxy clusters, the largest structures in the universe held together by gravity, are dynamic environments containing individual galaxies and huge amounts of hot gas and dark matter. Often, an enormous black hole in the center of a cluster can help drive its behavior. In the galaxy cluster Abell 2597, a giant central supermassive black hole is driving the gas outward and creating bubbles, or voids, within it. This composite image of Abell 2597 includes X-rays from Chandra (blue), optical data from the Digitized Sky Survey (orange), and emission from hydrogen atoms in optical light from the Las Campanas Observatory in Chile (red).

NASA's Marshall Space Flight Center manages the Chandra program. The Smithsonian Astrophysical Observatory's Chandra X-ray Center controls science operations from Cambridge, Massachusetts, and flight operations from Burlington, Massachusetts.



Kid Spot Jokes:

- ♦ **What is a light year?**
Same as a regular year but with less calories.
- ♦ **How did the astronaut serve drinks?**
In Sun glasses
- ♦ **How does the Solar system hold up its pants?**
With the Asteroid belt
- ♦ **When do astronauts have lunch?**
At launch time

Hubble Examines a Star-Forming Chamaeleon

Credit: NASA



Credit: NASA, ESA, K. Luhman and T. Esplin (Pennsylvania State University), et al., and ESO; Processing: Gladys Kober (NASA/Catholic University of America)

This NASA Hubble Space Telescope image captures one of three segments that comprise a 65-light-year wide star-forming region named the Chamaeleon Cloud Complex. The segment in this Hubble composite image, called Chamaeleon Cloud I (Cha I), reveals dusty-dark clouds where stars are forming, dazzling reflection nebulae glowing by the light of bright-blue young stars, and radiant knots called Herbig-Haro objects.

Herbig-Haro objects are bright clumps and arcs of interstellar gas shocked and energized by jets expelled from infant “protostars” in the process of forming. The white-orange cloud at the bottom of the image hosts one of these protostars at its center. Its brilliant white jets of hot gas are ejected in narrow torrents from the protostar’s poles, creating

the Herbig-Haro object HH 909A.

The cross-like spikes around bright stars in the image occur when light waves from a very bright point source (like a star) bend around Hubble’s cross-shaped struts that support the telescope’s secondary mirror. As the light waves pass these struts, they coalesce on the other side, creating the bright, spikey starburst effect we see.

Hubble studied Cha I as part of a search for extremely dim, low-mass brown dwarfs. These “failed stars” lie somewhere in size between a large planet and a small star (10 to 90 times the mass of Jupiter), and do not have enough mass to ignite and sustain nuclear fusion in their cores. Hubble’s search found six new low-mass brown dwarf candidates that are helping astronomers better understand these objects.

This 315-million-pixel composite image is comprised of 23 observations made by Hubble’s Advanced Camera for Surveys. Gaps between those observations were filled by 20 Wide Field and Planetary Camera 2 images. Any remaining gaps were filled with ground-based data from ESO’s VISTA VIRCAM

From the Board of Directors

Announcements

- ◇ San Jose Astronomical Association is monitoring closely the coronavirus, COVID-19, situation in Santa Clara county. In accordance with advisement from Santa Clara County Health Department we may be canceling some events in the next few weeks, as appropriate - especially those that involve close contact and shared surface on telescopes. Please check our Meetup to determine if an event has been canceled before heading out to the event. The board will periodically review the status of the outbreak and the guidance from the Santa Clara county to determine the impact upon upcoming events.
- ◇ Wolf Witt, Rob Jaworski, and Glenn Newell were reelected into their respective Board of Director positions. Steven Nelson was elected to the board.
- ◇ Officer elections for President, Vice-President, Treasurer, and Secretary positions will be held online via Electionbuddy and the new officers will be announced at the March Board meeting on March 19. Any member interested in running for these positions should reach out to the Secretary at secretary@sjaa.net for eligibility criteria and other details by March 8.
- ◇ The SJAA has a number of public outreach Programs which have been on hold due to the pandemic. We are looking to resume some of these Programs very soon, but we need volunteers. If you are able to volunteer to support the Starry Nights Program (held at Ranch Canada del Oro), please send an e-mail to Joe Fragola at starrynights@sjaa.net. He will contact you with further information.

Board Meeting Excerpts

December 18, 2021

In attendance

Swami, Sandy, Glenn, Ken, Peter, Wolf, Vini, Rob
Absent: Sukhada
Guests: Leslie, Joe, Steve

SJAA and West Valley College

SJAA is in discussions with West Valley College to check on the usage of their planetarium hall for SJAA meetings. Star parties on campus were also discussed. Further discussions are to be had on staffing issues.

SJAA Calendar addition

Guest Steve Nelson joined the meeting and suggested to add Oct. 1, 2022 as the 'International Observe the Moon' event to the SJAA calendar. Swami moved the motion to approve the 2022 calendar and it was seconded by Vini

SJAA In-Town Public Star Party

The Board agreed unanimously to do a Phase 1 minimum - EAA (Electronically Assisted Astronomy) event with gear like laser pointers, two eVscopes, QR codes/stands and a computer set-up.

January 15, 2022

In attendance

Swami, Vini, Wolf, Vini, Rob J, Ken, Peter
Absences: Glenn, Sukhada
Guests: Joe, Kanch, Leslie, Sheena

Enforcing existing policy regarding minors

The board had a discussion on the importance of enforcing the existing SJAA policy of requiring any minor in attendance at any SJAA function to be accompanied by a parent or legal guardian.

SJAA In-Town Public Star Party

Planning for a phase 1 minimum-EAA in-town public star party is in progress. The board discussed COVID procedures, emphasizing that Covid protocols continue to be effective.

February 19, 2022

In attendance

Wolf, Vini, Rob J, Glenn, Steven, Ken, Swami,
Absences: None
Guests: Kanch, Emi, Leslie

Board member Elections

Board member election results for Director 2, Director 4, Director 6, and Director 8 were announced during the February 19 board meeting.

SJAA Award Winner

Ken announced that Peter Melhus was this year's SJAA Award Winner. He has served as interim secretary, assisted with the electoral voting process, and chaired the 24" Dob task group. For a photo of Peter please look at Page 2 of this edition.

New Volunteer Coordinator

Nancy Lieu has been named the new volunteer coordinator. She replaces Sheena Williams.

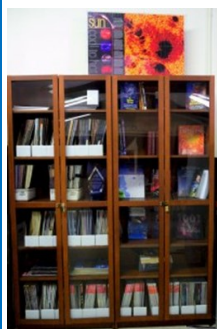
Funds for Solar Program

A decision was made to allocate funds to be used for equipment used in the solar program (with overlap to ITSP and other programs). Ken seconded the motion and it was a unanimous vote in favor.

Officer Elections

Officer elections for President, Vice President, Treasurer, and Secretary positions will be held online via Electionbuddy between March 10th and 16th and the new officers will be announced at the March 19. Anyone interested in running for an officer position needs to contact the secretary by March 8. Board meeting. Officer duties are described in Article 4 of Club By-Laws.
<https://www.sjaa.net/about/club-by-laws/#A4>.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The next Quick Start session during this time period is October 12th.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association

conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

SJAA Contacts

President/Dir:	Ken Miura
Vice President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Peter Melhus
Director:	Vini Carter
Director:	Steven Nelson
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Glenn Newell
Director:	Sukhada Palav
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Hands on Imaging:	Glenn Newell
Imaging SIG:	Hy Murveit
Intro to the Night Sky:	David Grover
In-Town Star Party:	Muditha Kanchana
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Donations:	Vacant
Quick START:	Vacant
Solar:	Wolf Witt
Starry Nights:	Joseph Fragola
School Events:	Vacant
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
Volunteer:	Nancy Lieu

E-mails: <http://www.sjaa.net/contact>

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

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Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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